

Klart' Digi

INTERNSHIP REPORT

Submitted by

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BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

CHETTINAD COLLEGE OF ENGINEERING & TECHNOLOGY

KARUR

Acknowledgement:

I would like to express my sincere gratitude to **Klart'Digi** for providing me with the opportunity to undergo my internship as a **Full Stack Developer Intern**. This internship has been a valuable learning experience that helped me gain practical exposure to real-world software development practices.

I express my heartfelt thanks to the management and technical team of **Klart'Digi** for their guidance, support, and encouragement throughout the internship period. Their mentorship helped me understand industry standards, project workflows, and professional responsibilities.

I also extend my sincere gratitude to the **Principal, Head of the Department**, and all the teaching and non-teaching staff members of **Chettinad College of Engineering & Technology** for their continuous support and motivation, which made this internship possible and grateful to everyone who directly or indirectly contributed to the successful completion of my internship.

CHETTINAD COLLEGE OF ENGINEERING & TECHNOLOGY

BONAFIDE CERTIFICATE

Certified that this internship report titled “**KLART’DIGI – FULL STACK DEVELOPMENT**” is the bonafide work of **Mr. Dhanush Kumar R**, who has successfully completed his internship as a **Full Stack Developer Intern** at **Klart’Digi** from **21.07.2025 to 10.12.2025**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering**.

Mentor

Placement In Charge

HOD

Principal

Introduction:

Klart'Digi is a technology-driven organization focused on delivering innovative and scalable digital solutions. The company specializes in developing modern web applications that address real-world business and institutional needs using cutting-edge technologies.

In today's rapidly evolving digital landscape, organizations require reliable and scalable software systems to manage data efficiently and streamline operations. Klart'Digi bridges this gap by designing and implementing robust full-stack solutions that align with industry standards.

The organization emphasizes practical learning, real-time project exposure, and professional development, enabling interns and developers to gain hands-on experience with modern development tools, frameworks, and best practices.

Our Expertise Includes:

- **Real-World Projects:** Providing hands-on experience by working on realtime industrial challenges.
- **Industry-Ready Skills:** Offering training in cutting-edge technologies and methodologies to meet industry standards.
- **Practical Learning Environment:** Simulating workplace scenarios to bridge the transition from academia to industry.
- **Soft Skills Development:** Enhancing communication, teamwork, and problem-solving abilities for holistic growth.
- **Career Readiness:** Preparing students to succeed professionally with a focus on both technical and interpersonal skills.

Designation:

During my internship at **Klart'Digi**, I worked as a **Full Stack Developer Intern**. The internship duration was from **21st July 2025 to 10th December 2025**.

Technologies Used

- Frontend: **Angular**
- Backend: **Node.js, Express.js**
- Databases: **MySQL, MongoDB**
- Tools: **Git, Postman**

Project Worked On School Management System – Klartopedia

The primary objective of this project was to develop a comprehensive school management portal that simplifies academic and administrative operations. The system includes modules for managing students, staff, classes, attendance, and academic records through a centralized web application.

Work Report:

Work Carried Out:

During my internship at Klart'Digi under the guidance of Mr.Vijay Veeraselvan and team.

Month 1 (July 2025 – August 2025)

Project Understanding, Requirement Analysis & System Design

During the first month of my internship at Klart'Digi, I focused on understanding the organization, project scope, and system requirements of the School Management System – Klartopedia.

Activities Performed:

- Attended orientation sessions to understand company policies, development workflow, and project lifecycle.
- Studied the objectives and scope of the School Management System, which aims to digitalize academic and administrative processes.
- Analyzed functional and non-functional requirements of the system.
- Understood the three-tier architecture (Frontend, Backend, Database).
- Set up development environments for Angular, Node.js, and Express.js.
- Learned REST API architecture and request-response mechanisms.
- Explored MySQL for relational data and MongoDB for flexible document storage.
- Practiced API testing and debugging using Postman.
- Used Git for source code version control.

Modules Studied:

- User Authentication Module
- Student Management Module
- Staff Management Module
- Class & Section Management Module

Learning Outcome:

- Clear understanding of real-time project planning and system architecture.
- Improved knowledge of backend-frontend interaction.
- Gained experience in professional development tools and workflows.

Month 2 (September 2025)

Frontend Development & Core Module Implementation

In the second month, I actively contributed to frontend and backend development of core modules of the Klartopedia system.

Activities Performed:

- Designed and developed Angular components for multiple system modules.
- Implemented responsive forms with validations.
- Integrated frontend components with backend REST APIs.
- Developed CRUD operations using Node.js and Express.js.
- Connected APIs with MySQL and MongoDB databases.
- Used Postman to test API responses and error handling.
- Fixed UI bugs and data-binding issues.
- Participated in team discussions and sprint-based development.

Modules Implemented:

- Student Management Module
 - Student registration and profile management
 - Academic details and personal information handling
- Staff Management Module
 - Teacher and non-teaching staff records
 - Department and designation mapping
- Class & Section Module
 - Class creation and section allocation
 - Mapping students and staff to classes

Learning Outcome:

- Gained hands-on experience in Angular component architecture.
- Understood full stack integration in real-time applications.
- Improved debugging and collaboration skills.

Month 3 (October 2025)

Advanced Modules, Authentication & Performance Optimization

During this phase, the focus was on advanced features, security, and system optimization.

Activities Performed:

- Implemented authentication and authorization mechanisms.
- Developed role-based access control for Admin, Teacher, and Student roles.
- Optimized backend APIs for better performance.
- Refactored Angular components to improve maintainability.

- Implemented proper exception handling and validations.
- Assisted in optimizing database queries.
- Participated in code reviews and followed coding standards.

Modules Implemented:

- User Authentication Module
 - Login and logout functionality
 - Role-based access control
- Attendance Management Module
 - Daily attendance tracking
 - Class-wise and student-wise attendance records
- Academic Management Module
 - Subject allocation
 - Academic year and term management

Learning Outcome:

- Improved understanding of application security.
- Learned performance optimization techniques.
- Gained exposure to scalable software development practices.

Month 4 (November 2025 – December 2025)

- ***Testing, Integration, Documentation & Deployment Readiness***
- The final month focused on system testing, stabilization, and preparation for deployment.
- **Activities Performed:**
 - Performed end-to-end testing of all implemented modules.
 - Identified and fixed bugs related to data flow and UI issues.
 - Integrated multiple modules into a unified system.
 - Verified role-based access across modules.
 - Ensured frontend and backend validation consistency.
 - Prepared technical documentation for system usage.
 - Learned basic deployment and production workflows.
 - Assisted in final review and feedback implementation.
- **Modules Finalized:**
 - Examination & Result Module
 - Exam scheduling
 - Result entry and report generation
 - Notification Module
 - System alerts and academic notifications
 - Reports Module

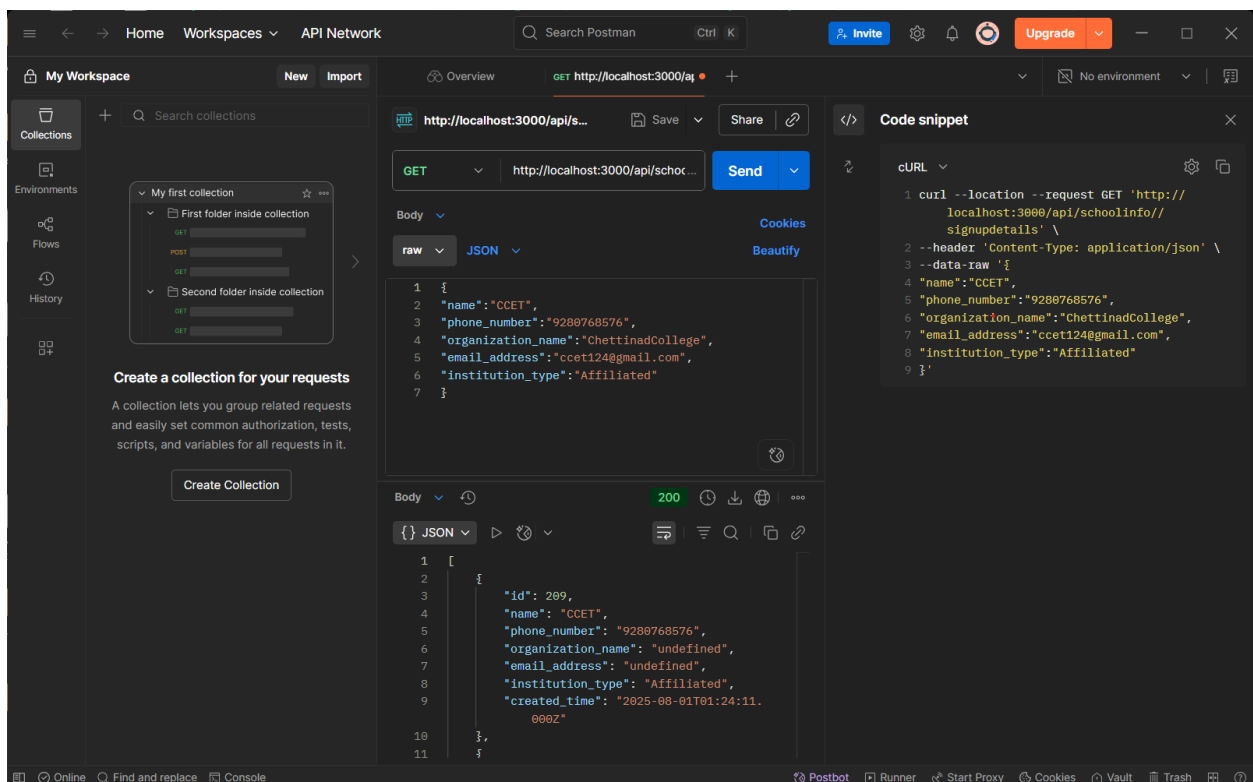
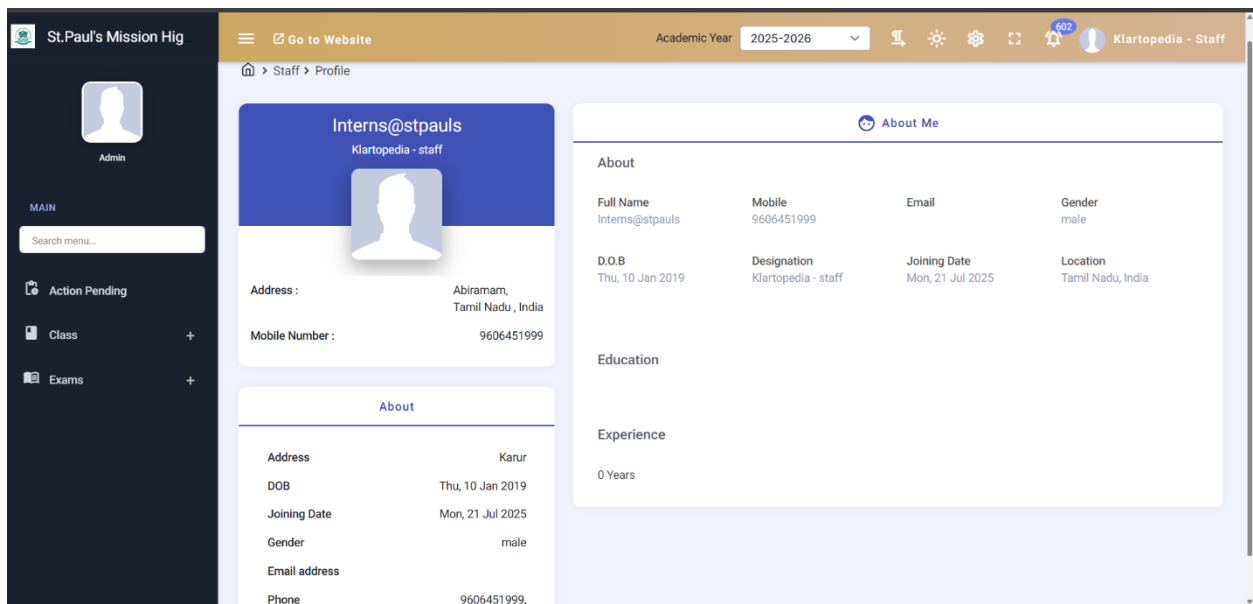
- Student, staff, and attendance reports
- **Learning Outcome:**
 - Understood software testing methodologies.
 - Learned system integration techniques.
 - Gained insight into real-world deployment readiness.

The screenshot shows the St. Paul's Mission Hg dashboard. The left sidebar contains a user profile for 'Admin' and a 'MAIN' menu with options: 'Action Pending', 'Class' (selected), 'All Class', and 'Exams'. The top header includes a 'Go to Website' button, 'Academic Year 2025-2026', and user information 'Klartopedia - Staff'. The main content area displays 'Classes > All Classes' with a table of class details. A notifications overlay is visible on the right, listing updates for Mahadevaswamy, Amitha Susan Mathew, SANTHOSH KUMAR, Tejas L, and Reshma Banu.

All Class	S.No	Class Name	Class Teacher	Board Name	Display Order	Capacity
☐	1	LKG	Anitha G	stateboardenglish	0	43
☐	2	UKG	Jabeena Taj	stateboardenglish	1	36
☐	3	1st STD A Section	Vedashree	stateboardenglish	2	29
☐	4	1st STD B Section	Meghana N	stateboardenglish	3	30
☐	5	2nd STD A Section	Pushpalatha A	stateboardenglish	4	37
☐	6	2nd STD B Section	Helen J	stateboardenglish	5	39
☐	7	3rd STD	Komala	stateboardenglish	6	49
☐	8	4th STD	Shashikala	stateboardenglish	7	36
☐	9	5th STD A Section	Ravichandra H R	stateboardenglish	8	25

This screenshot shows the same dashboard as above, but with an additional 'Actions' column in the class table. The 'Actions' column contains icons for viewing details, editing, and deleting each class entry.

All Class	S.No	Class Name	Class Teacher	Board Name	Display Order	Capacity	Actions
☐	1	LKG	Anitha G	stateboardenglish	0	43	View Edit Delete
☐	2	UKG	Jabeena Taj	stateboardenglish	1	36	View Edit Delete
☐	3	1st STD A Section	Vedashree	stateboardenglish	2	29	View Edit Delete
☐	4	1st STD B Section	Meghana N	stateboardenglish	3	30	View Edit Delete
☐	5	2nd STD A Section	Pushpalatha A	stateboardenglish	4	37	View Edit Delete
☐	6	2nd STD B Section	Helen J	stateboardenglish	5	39	View Edit Delete
☐	7	3rd STD	Komala	stateboardenglish	6	49	View Edit Delete
☐	8	4th STD	Shashikala	stateboardenglish	7	36	View Edit Delete
☐	9	5th STD A Section	Ravichandra H R	stateboardenglish	8	25	View Edit Delete



Conclusion:

The internship at **Klart'Digi** as a **Full Stack Developer Intern** was an enriching and valuable experience. It enhanced my technical knowledge in full-stack development and improved my problem-solving, teamwork, and professional skills.

Working on the **School Management System – Klartopedia** provided real-world exposure to building scalable web applications using modern technologies. This internship has strengthened my foundation and prepared me to face future challenges in the software industry with confidence.

10th Dec 2025

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that Mr. Dhanush Kumar R, a student of Chettinad College of Engineering, has successfully completed his internship with Klart'Digi as a Full Stack Developer Intern.

His internship was for a duration from 21st July 2025 to 10th December 2025. During this period, he was involved in full stack application development activities and contributed effectively to assigned project tasks.

We found his performance during the internship to be very good. He demonstrated technical competence, dedication, and a professional attitude throughout his tenure.

Email ID: trdhanush1234@gmail.com

We wish him the very best for his future career and academic pursuits.




Authorized Signatory

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